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The Editors, *Nature Communications*

Dear Editors,

We are pleased to submit our manuscript, "Optimal stimulation sites are not the most affected: personalised network models of Alzheimer's disease," by Cristiano Capone, Enza Cece, Andrea Ciardiello, Guido Gigante, Evaristo Cisbani and Maurizio Mattia, for consideration as an Article in *Nature Communications*.

Altered brain-network dynamics can be corrected in two very different ways, by reshaping the connectivity *graph* or by injecting focal *activity* at a node, and these interventions need not point to the same target. Testing this causally in Alzheimer's disease, widely regarded as a distributed network disorder, requires an individualised generative model that can be perturbed *in silico*. We fit subject-specific, cross-subject-identifiable reservoir-computing models of each patient's lagged functional connectivity and use them as a testbed for such counterfactual interventions.

On the fitted model the two interventions behave very differently. A graph correction that restores control-like dynamics is intrinsically distributed, a coordinated multi-site change of the connectivity kernel that a single-node edit cannot reproduce, whereas a focal activity drive is effective from one physically realisable site. Crucially, that optimal site is *not* where the graph is most altered: the most graph-changed nodes (weakly-connected subcortical regions) and the effective stimulation sites (well-connected cortical regions) are largely disjoint. The dissociation follows from connectivity itself, the most affected nodes are the most weakly coupled and hence poor actuators, so the target must be chosen by its effect on the dynamics, not by where the graph change concentrates. A real-time closed-loop controller then reaches comparable efficacy at substantially lower dose using only causally available information.

These results recast a central question in targeted neuromodulation, "where should we stimulate?", as one about the interaction between intervention modality and network connectivity, answerable only with an individualised, perturbable model. We believe this will interest the broad readership of *Nature Communications* across network neuroscience, computational modelling and neuromodulation, beyond Alzheimer's disease.

The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission. The data derive from the Alzheimer's Disease Neuroimaging Initiative and are used under its Data Use Agreement. We declare no competing interests, and would gladly suggest suitable referees on request. We thank you for considering our work.

Sincerely,

Cristiano Capone, on behalf of all authors